

## Module A (Lec 5–6): Steepest Descent Analysis on Quadratics

**The Setup.** Minimize  $f(x) = \frac{1}{2}x^\top Qx - b^\top x$  where  $Q \succ 0$ .

• **Algorithm:**  $x^{(k+1)} = x^{(k)} - \alpha_k g^{(k)}$ , with  $g^{(k)} = \nabla f(x^{(k)})$ .

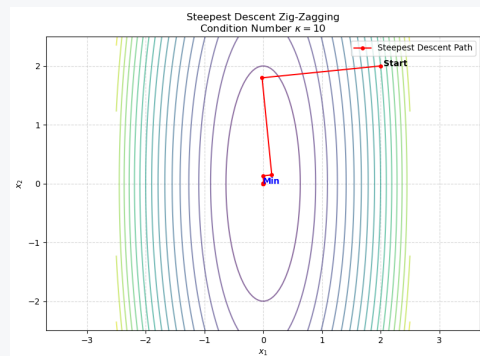
• **Exact Line Search:**  $\alpha_k = \frac{g^{(k)\top} g^{(k)}}{g^{(k)\top} Q g^{(k)}}$ .

**Convergence Rate (Theorem 8.2).** Let  $V(x) = f(x) - f(x^*)$ .

$$V(x^{(k+1)}) \leq \left( \frac{\lambda_{\max}(Q) - \lambda_{\min}(Q)}{\lambda_{\max}(Q)} \right) V(x^{(k)}).$$

• **Condition Number**  $\kappa(Q) = \lambda_{\max}/\lambda_{\min}$  dictates speed.

•  $\kappa \gg 1 \implies$  Slow, **Orthogonal (Why?) Zig-zagging**  $g^{(k+1)} \perp g^{(k)}$ .



## Practice: Exercise 8.8 (Fixed-Step Gradient Method)

Largest stepsize range for global convergence Consider the function

$$f(x) = 3(x_1^2 + x_2^2) + 4x_1x_2 + 5x_1 + 6x_2 + 7, \quad x = [x_1, x_2]^\top \in \mathbb{R}^2.$$

Suppose that we use a fixed-step-size gradient algorithm to find the minimizer of  $f$ :

$$x^{(k+1)} = x^{(k)} - \alpha \nabla f(x^{(k)}).$$

Find the largest range of values of  $\alpha$  for which the algorithm is globally convergent.

(a) **Quadratic structure.** Compute  $\nabla f(x)$  and the Hessian  $H = \nabla^2 f(x)$ .

**Answer:**  $\nabla f(x) = \begin{bmatrix} 6x_1 + 4x_2 + 5 \\ 4x_1 + 6x_2 + 6 \end{bmatrix}$  and  $H = \begin{bmatrix} 6 & 4 \\ 4 & 6 \end{bmatrix} \succ 0$  (eigenvalues 2, 10  $> 0$ ).

(b) **Eigenvalues & Positive Definiteness.** Compute the eigenvalues of  $H$ . Is  $H$  positive definite?

**Answer:**  $\lambda_{\min} = 2$ ,  $\lambda_{\max} = 10$ . Yes, all  $\lambda_i > 0$ .

(c) **Error Iteration.** Prove  $e^{(k)} := x^{(k)} - x^* \implies e^{(k+1)} = (I - \alpha H)e^{(k)}$ .

**Answer:** Since  $f$  is quadratic,  $\nabla f(x) = Hx + c$ .

Minimizer  $x^*$  satisfies  $\nabla f(x^*) = Hx^* + c = 0$ .

Gradient at step  $k$ :  $\nabla f(x^{(k)}) = Hx^{(k)} + c = Hx^{(k)} - Hx^* = H(x^{(k)} - x^*)$ .

Algorithm update:  $x^{(k+1)} = x^{(k)} - \alpha \nabla f(x^{(k)}) = x^{(k)} - \alpha H(x^{(k)} - x^*)$ .

Subtract  $x^*$ :  $x^{(k+1)} - x^* = (x^{(k)} - x^*) - \alpha H(x^{(k)} - x^*) = (I - \alpha H)e^{(k)}$ .

(d) **Global convergence condition.** Using  $e^{(k)} := x^{(k)} - x^*$  and  $e^{(k+1)} = (I - \alpha H)e^{(k)}$ , find the largest range of  $\alpha$  such that  $\rho(I - \alpha H) < 1$ .

**Answer:** Need  $|1 - \alpha \lambda_i| < 1$  for  $i = 1, 2$ . Since  $\lambda_i > 0$ , this is equivalent to  $0 < \alpha < \frac{2}{\lambda_{\max}} = \frac{2}{10} = \frac{1}{5}$ .

## Module B (Week 3 & Chapter 9): Newton's Method in $\mathbb{R}^n$

**The Idea.** Locally approximate  $f$  with a **quadratic** Taylor expansion and jump to its minimum.

$$x^{(k+1)} = x^{(k)} - \frac{f'(x^{(k)})}{f''(x^{(k)})} \implies x^{(k+1)} = x^{(k)} - [\nabla^2 f(x^{(k)})]^{-1} \nabla f(x^{(k)}).$$

Let  $H_k = \nabla^2 f(x^{(k)})$  and  $g_k = \nabla f(x^{(k)})$ . The step solves the linear system  $H_k d_k = -g_k$ .

**Properties.**

- **Quadratic convergence** ( $p = 2$ ) near  $x^*$ .
- Finds the minimum of a **perfect quadratic** in **1 step**.
- **Pitfall:** If  $H_k \not\succeq 0$ , the step might not be a descent direction (attracts to saddle points).

### Practice: Exercise 9.1 (Newton's Method)

**$f(x) = (x - x_0)^4$ : convergence and order:** Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be given by  $f(x) = (x - x_0)^4$ , where  $x_0 \in \mathbb{R}$  is a constant. Suppose that we apply Newton's method to the problem of minimizing  $f$ , with iterates  $x^{(0)}, x^{(1)}, x^{(2)}, \dots$

(a) Write down the update equation for Newton's method applied to the problem.

**Answer:**  $f'(x) = 4(x - x_0)^3$ ,  $f''(x) = 12(x - x_0)^2$ . For  $x^{(k)} \neq x_0$ ,  $x^{(k+1)} = x^{(k)} - \frac{f'(x^{(k)})}{f''(x^{(k)})} = x^{(k)} - \frac{x^{(k)} - x_0}{3} = \frac{2}{3}x^{(k)} + \frac{1}{3}x_0$ . (If  $x^{(k)} = x_0$ , we are already at the minimizer.)

(b) Let  $y^{(k)} = |x^{(k)} - x_0|$ . Show that the sequence  $\{y^{(k)}\}$  satisfies  $y^{(k+1)} = \frac{2}{3}y^{(k)}$ .

**Answer:**  $x^{(k+1)} - x_0 = \frac{2}{3}(x^{(k)} - x_0) \Rightarrow y^{(k+1)} = \left|\frac{2}{3}(x^{(k)} - x_0)\right| = \frac{2}{3}y^{(k)}$ .

(c) Show that  $x^{(k)} \rightarrow x_0$  for any initial guess  $x^{(0)}$ .

**Answer:** From (b):  $y^{(k)} = \left(\frac{2}{3}\right)^k y^{(0)} \rightarrow 0$ , hence  $|x^{(k)} - x_0| \rightarrow 0$  so  $x^{(k)} \rightarrow x_0$ .

(d) Show that the order of convergence of the sequence  $\{x^{(k)}\}$  in part (b) is 1.

**Answer:**  $\frac{|x^{(k+1)} - x_0|}{|x^{(k)} - x_0|} = \frac{y^{(k+1)}}{y^{(k)}} = \frac{2}{3} \rightarrow \frac{2}{3} \neq 0$ , so convergence is linear (order  $p = 1$ ) with asymptotic constant  $2/3$ .

(e) Theorem 9.1 states that under certain conditions, the order of convergence of Newton's method is at least 2. Why does that theorem not hold in this particular problem?

**Answer:** A key condition is that the second derivative at the minimizer is nonzero (equivalently, the Hessian is nonsingular). Here  $f''(x_0) = 12(x_0 - x_0)^2 = 0$ , so the required nondegeneracy fails; the local quadratic-convergence guarantee does not apply.

### Pitfalls

- **Gradient Descent:** "Steepest"  $\neq$  "Straight to minimum" (unless  $\kappa = 1$ ). It often zig-zags.
- **Newton's Method:** Expensive for large  $n$  (requires  $H^{-1}$ ).
- **Global vs Local:** Both methods find *local* minimizers (or stationary points).